

Sophia Lin

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Education

Harvard University

Expected May 2029

A.B. Candidate, Computer Science and Statistics; Advanced Standing

Current coursework: Reinforcement Learning, Theoretical Computer Science, Effective AI Support in Human-AI Settings (graduate), Topics at the Interface of Computer Science and Economics (graduate)

Selected coursework: Data Structures and Algorithms; Probability, Statistical Inference, Linear Algebra and Real Analysis, Critical Thinking in Data Science (graduate)

Thomas Jefferson High School for Science and Technology

June 2025

Selected Research and Technical Projects

National Institute of Standards and Technology (NIST), Information Technology Laboratory

Undergraduate Researcher, NIST PREP

May 2026 – Present

Adversarial Verification for Agentic AI Grounding

Project Lead; Ongoing

- Designing controlled experiments testing whether independent, competing-objective verifier agents improve the factual grounding of agentic outputs
- Building reproducible Python evaluation tooling for stress-testing outputs against novel adversarial training scenarios testing citation-grounded reasoning, deception, hallucination, contradiction handling, and reliance on retrieved evidence over conflicting parametric knowledge

Characterizing LLM Judges as Measurement Instruments

Project Lead

- Led the end-to-end design of a measurement framework evaluating whether LLM judges reliably assess entailment, claim reasoning and decomposition, error recognition, confidence, and abstention
- Designed 15+ evaluation probes spanning blind, correct-guidance, and deliberately false-guidance conditions to distinguish genuine evaluation capability from prompt sensitivity and sycophancy

Research Science Institute, MIT CSAIL and Harvard Medical School

Machine Learning Research Scholar

June – August 2024

Bayesian Modeling of Multi-Stage Censoring

Project Co-Lead

- Co-developed a Bayesian model of sequential selection and censoring that recovered latent patient risk from incomplete clinical decision pipelines and revealed disparities in triage

Stony Brook University

AI Researcher

May 2023 – March 2024

FRISTS: Interpretable Time-Series Prediction from Clinical Records

Project Lead

- Led development of an interpretable prediction model that extracted temporal patterns and produced human-auditable risk estimates. Built and validated the method on 18 million longitudinal health records, outperforming prior heart-failure prediction baselines

U.S. Government Accountability Office, Applied Research and Methods

Data Science Intern

June – August 2025

Longitudinal Analysis of Federal ROTC Programs

Contributor

- Built Python, R, and SAS pipelines integrating 30 years of enrollment, commissioning, funding, and program data; implemented edit checks and clustering analyses across inconsistent longitudinal records

Selected Publications

FRISTS: High-Performing Interpretable Medical Prediction

S. Lin, X. Dong, F. Wang

IEEE BigData, 2024 — Oral Presentation. [DOI](#)

A Bayesian Model for Multi-stage Censoring

S. Sadhuka, S. Lin, B. Berger, E. Pierson

ML4H Symposium, 2025 — Spotlight Talk. [arXiv](#)

Hybrid AI Framework for Remote Patient Heart Failure Risk Prediction

W. Xu, J. Zhao, K. Parvataneni, V. Zhang, S. Lin, H. Wang

2022 Global Reliability and Prognostics and Health Management Conference (PHM-Yantai) — Best Paper. [DOI](#)